

Multiple Testing Correction

Preserving the Meaning of Statistical Evidence

StatHarbor Analytics

Course Map

Foundations of Hypothesis Testing

The Multiplicity Problem

Error Criteria and Their Meanings

Classical FWER Procedures

FDR Procedures and Their Theory

Resampling and Dependence-Aware Procedures

Multiple Comparisons in Models and Designs

Why This Topic Matters

Central Question

When many hypotheses are tested, what probability statement does “significant” still support?

- Classical testing attaches a Type I error guarantee to a specified decision rule.
- Repeating individual tests creates a [joint decision problem](#).
- Multiple testing is therefore about [simultaneous inference](#), not merely about adjusting p -values.

A Minimal Decision-Theoretic View

For one hypothesis H_0 against H_1 , a test is a rule $\varphi(X) \in \{0, 1\}$, where

$$\varphi(X) = 1 \quad \text{means reject } H_0.$$

Level- α Test

A test is level α if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\varphi(X) = 1) \leq \alpha.$$

- The guarantee is attached to a **decision rule**.
- The null may be simple or composite.
- The rejection probability is controlled **uniformly over the null model**.

Type I Error, Type II Error, and Power

	Fail to Reject H_0	Reject H_0
H_0 true	Correct	Type I error
H_0 false	Type II error	Correct (power)

- Type I error control is a property of the test under the null model.
- Power is a property of the test under alternatives.
- Multiple testing preserves this tension but changes the error criterion from a single decision to a family of decisions.

What a p -value Is

Suppose a test statistic $T(X)$ is chosen so that large values are less compatible with H_0 .

General Definition

The p -value is

$$p(x) = \sup_{\theta \in \Theta_0} \mathbb{P}_{\theta}(T(X) \geq T(x)),$$

for a one-sided upper-tail test, with the ordering changed appropriately for other rejection regions.

- A p -value is a function of the observed data and the null model.
- It measures **null-model extremeness**, not practical importance.
- It is valid only relative to the test and model from which it was constructed.

What a p -value Is Not

Common Misinterpretations

A p -value is not:

- the probability that H_0 is true,
- the probability that the finding is a false positive,
- the probability that the result replicates,
- a measure of the size or relevance of the effect.

Safe Interpretation

A small p -value means the observed data would be unusual under the null model and the chosen test statistic.

The Individual-Test Guarantee

Level- α Inference for One Decision

For a pre-specified test,

$$\mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}) \leq \alpha.$$

- This is an individual-test guarantee.
- It does not automatically imply control over a collection of many decisions.
- The issue in multiple testing is not that inference “breaks”; the issue is that the error target changes.

From Individual Tests to Simultaneous Inference

Key Principle

Classical testing theory remains valid under multiplicity, but the relevant probability statement must be defined over the **entire family of decisions**.

- Single hypothesis $\rightarrow$ control one Type I error probability.
- Multiple hypotheses $\rightarrow$ control a simultaneous error criterion such as FWER or FDR.

Interpretation

Multiplicity is a shift from **individual error control** to **family-level error control**.

Why “Preserving the Meaning of Evidence” Is the Right Framing

If the same nominal threshold α is used repeatedly,

- the per-test statement remains correct,
- but the family-level interpretation becomes misleading.

Consequence

Without simultaneous control, “significant” may reflect the number of opportunities for chance rather than the strength of signal.

Multiplicity Arises in Many Forms

Explicit Multiplicity

- multiple endpoints,
- multiple treatment arms,
- all pairwise comparisons,
- voxel-wise or gene-wise testing.

Implicit Multiplicity

- subgroup searches,
- model selection,
- repeated transformations,
- optional stopping,
- specification searching.

A Simple Global-Null Calculation

Suppose m tests are carried out independently and all null hypotheses are true. Then

$$\mathbb{P}(\text{no false rejection}) = (1 - \alpha)^m.$$

Therefore

$$\mathbb{P}(\text{at least one false rejection}) = 1 - (1 - \alpha)^m.$$

- For $m = 1$ and $\alpha = 0.05$, the error probability is 0.05.
- For $m = 20$, it becomes $1 - 0.95^{20} \approx 0.642$.
- For $m = 100$, it is about 0.994.

The Global Null Under Dependence

Independence gives an exact formula, but the phenomenon does not rely on independence.

Union Bound

For any events $A_1, \dots, A_m$,

$$\mathbb{P}\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m \mathbb{P}(A_i).$$

Applying this with $A_i = \{p_i \leq \alpha\}$ gives

$$\mathbb{P}(\text{at least one false rejection}) \leq m\alpha,$$

which already shows that repeated testing inflates the opportunity for error.

Illustration of Error Inflation

Number of tests m	$1 - (1 - 0.05)^m$
1	0.050
5	0.226
10	0.401
20	0.642
50	0.923
100	0.994

Interpretation

At fixed nominal level, repeated testing makes false rejections expected rather than exceptional.

The Jelly Bean Lesson

The famous “jelly beans cause acne” cartoon captures the essence of multiplicity.

- Twenty colors are tested separately.
- Each test is valid at level 0.05.
- One “significant” result appears by chance.

Lesson

A significant finding among many tested hypotheses is not self-authenticating; it must be interpreted relative to the family of questions asked.

The Dead Salmon Lesson

Massively parallel settings make the issue unavoidable.

- In neuroimaging, one scan may involve tens or hundreds of thousands of voxel-wise tests.
- Without correction, some voxels will appear active even when no signal exists.

Lesson

Large-scale testing turns multiplicity from a methodological subtlety into a first-order inferential problem.

Implicit Multiplicity and the Garden of Forking Paths

Even if only one final analysis is reported, multiplicity may have been introduced through analytical flexibility.

- trying several models,
- changing covariate sets,
- choosing one among multiple endpoints,
- inspecting data before locking the analysis.

Key Point

The family of potential decisions matters, not merely the single decision eventually written in the paper.

A Formal Counting Framework

For m tested hypotheses, write:

	Declared null	Declared signal	Total
True null	U	V	m_0
False null	T	S	$m - m_0$
Total	$m - R$	R	m

Here

$$R = S + V, \quad m_0 = U + V.$$

Most simultaneous error criteria are functions of V and R .

What Needs To Be Controlled?

The key question is no longer “what is the error rate of this one test?”

Simultaneous Error Criteria

Common targets include

FWER, FDR, k -FWER, PFER, and related quantities.

The right target depends on the scientific objective:

- confirmatory testing,
- screening and discovery,
- ranking and prioritization,
- adaptive decision pipelines.

Family-Wise Error Rate

Definition

$$\text{FWER} = \mathbb{P}(V \geq 1).$$

- It is the probability of **at least one false rejection**.
- It is the natural criterion when any false claim is costly.
- It is common in confirmatory clinical trials and high-stakes regulatory settings.

Weak and Strong Control

Weak Control

A procedure controls FWER weakly at level α if

$$\text{FWER} \leq \alpha$$

under the global null $m_0 = m$.

Strong Control

A procedure controls FWER strongly at level α if

$$\text{FWER} \leq \alpha$$

for every configuration of true and false null hypotheses.

Why Strong Control Matters

Suppose some hypotheses are truly non-null and others are null.

- Weak control says nothing in this mixed configuration.
- Strong control still guarantees that the chance of at least one false rejection remains bounded.

Practical Implication

Confirmatory programs usually require strong control, because real alternatives do not excuse false positive claims on the remaining true nulls.

Per-Family Error Rate and k -FWER

Other criteria target different notions of false discovery burden.

Examples

$$\text{PFER} = \mathbb{E}(V), \quad k\text{-FWER} = \mathbb{P}(V \geq k).$$

- PFER controls the expected number of false rejections.
- k -FWER allows up to $k - 1$ false rejections before counting failure.
- These criteria can be useful when “at least one” is too strict.

False Discovery Rate

Definition

$$\text{FDR} = \mathbb{E} \left[\frac{V}{\max(R, 1)} \right].$$

- The random ratio $V / \max(R, 1)$ is the **false discovery proportion**.
- FDR controls the expected proportion of false discoveries among the rejected hypotheses.
- It is tailored to discovery-rich settings where some false positives may be tolerable.

Relationship Between FWER and FDR

Because

$$\frac{V}{\max(R, 1)} \leq \mathbf{1}(V \geq 1),$$

we have

$$\text{FDR} \leq \text{FWER}.$$

- If all nulls are true, then every rejection is false, so

$$\text{FDR} = \text{FWER}.$$

- When some alternatives are real, FDR becomes less stringent than FWER.

Positive FDR and Local FDR

Two related notions often appear in modern large-scale inference.

Definitions

$$\text{pFDR} = \mathbb{E} \left[\frac{V}{R} \mid R > 0 \right], \quad \text{lfdr}(z) = \mathbb{P}(H_0 \mid Z = z)$$

under a two-group mixture model for Z .

- pFDR conditions on making at least one rejection.
- lfdr is a model-based posterior quantity for an individual observed statistic.

How Error Criteria Change the Scientific Question

FWER Mindset

“Can I protect against *any* false claim?”

FDR Mindset

“Among my discoveries, can I control the *expected fraction* that are false?”

The choice is not only mathematical; it encodes the scientific tolerance for error and missed signal.

Bonferroni Correction

Procedure

Reject H_i only if

$$p_i \leq \frac{\alpha}{m}.$$

- Bonferroni is simple, general, and distribution-free at the family level.
- It does not require independence.
- It is often the first benchmark for simultaneous control.

Why Bonferroni Works

Let I_0 denote the set of true null hypotheses, with $|I_0| = m_0$. Then

$$\text{FWER} = \mathbb{P} \left(\bigcup_{i \in I_0} \left\{ p_i \leq \frac{\alpha}{m} \right\} \right) \leq \sum_{i \in I_0} \mathbb{P} \left(p_i \leq \frac{\alpha}{m} \right).$$

If each true-null p -value is stochastically no smaller than $\text{Uniform}(0, 1)$, then

$$\mathbb{P} \left(p_i \leq \frac{\alpha}{m} \right) \leq \frac{\alpha}{m},$$

so

$$\text{FWER} \leq m_0 \frac{\alpha}{m} \leq \alpha.$$

Bonferroni as a Worst-Case Bound

Bonferroni pays no attention to dependence or structure.

- This is why it is valid under arbitrary dependence.
- This is also why it can be conservative.

Interpretation

Bonferroni behaves as if each test is a fresh opportunity for error, with no credit given for shared information or correlation.

Šidák Correction

If the true-null tests are mutually independent,

$$\mathbb{P}(\text{no false rejection}) = (1 - \alpha_{\text{Šidák}})^{m_0}.$$

Setting this equal to $1 - \alpha$ yields

$$\alpha_{\text{Šidák}} = 1 - (1 - \alpha)^{1/m}.$$

- Slightly less conservative than Bonferroni.
- Exact under independence.
- Loses its exactness under arbitrary dependence.

Holm's Step-Down Procedure

Order the p -values:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}.$$

Then compare sequentially:

$$p_{(1)} \leq \frac{\alpha}{m}, \quad p_{(2)} \leq \frac{\alpha}{m-1}, \quad \cdots, \quad p_{(m)} \leq \alpha.$$

Stop at the first failure and retain all remaining hypotheses.

Why Holm Improves on Bonferroni

- Holm preserves strong FWER control under arbitrary dependence.
- It is **uniformly at least as powerful** as Bonferroni.
- Once the smallest p -values survive the strictest thresholds, later hypotheses face less severe penalties.

Interpretation

Holm recycles the family-level error budget more efficiently than simple Bonferroni splitting.

Hochberg's Step-Up Procedure

Again order the p -values. Starting from the largest, find the first index k such that

$$p_{(k)} \leq \frac{\alpha}{m - k + 1}.$$

Reject $H_{(1)}, \dots, H_{(k)}$.

- Often more powerful than Holm.
- Requires stronger conditions, typically independence or assumptions ensuring Simes' inequality.

Simes' Inequality and Why It Matters

Simes showed that under independence,

$$\mathbb{P} \left(\min_{1 \leq i \leq m} \frac{mp_{(i)}}{i} \leq \alpha \right) \leq \alpha.$$

- This inequality underlies both the Hochberg procedure and the logic behind BH.
- It connects ordered p -values to global and family-level control.

Closure and Consonance

A rigorous way to build strong FWER procedures is the closed testing principle.

Closed Testing

A hypothesis H_i is rejected only if every intersection hypothesis containing H_i is rejected by a valid level- α test.

- This yields strong control by construction.
- Many classical multiple testing procedures can be interpreted through closure.

Gatekeeping and Hierarchical Testing

In confirmatory programs, hypotheses may be ordered in families.

- Primary endpoints are tested first.
- Secondary families are tested only if the earlier family succeeds.
- This preserves the global error budget while encoding scientific priority.

Message

Multiplicity correction is often embedded in the [testing architecture](#), not just in a formula.

Tukey's HSD for All Pairwise Comparisons

In one-way ANOVA with homoscedastic errors, Tukey's method targets all pairwise contrasts.

- It is designed specifically for simultaneous inference on

$$\mu_i - \mu_j, \quad i < j.$$

- It uses the studentized range distribution.
- It gives exact family-wise control in the balanced homoscedastic setting.

Why Tukey Can Beat Bonferroni for Pairwise Means

Bonferroni ignores the correlation and geometry of pairwise comparisons.

- Tukey exploits the fact that all comparisons share a pooled variance estimate.
- The joint distribution of the maximum standardized mean difference is built into the procedure.

Interpretation

A structured problem often deserves a structured correction rather than a generic worst-case bound.

Dunnett's Procedure

When several treatments are each compared with a common control,

- Dunnett's procedure is more efficient than treating all possible comparisons equally,
- because it targets only the scientifically relevant contrasts,
- and uses their joint dependence structure.

Lesson

The best correction depends on the **comparison family** actually under study.

Scheffé and General Contrasts

Scheffé's method provides simultaneous confidence bounds for all contrasts in ANOVA.

- It is extremely general.
- It can be conservative when only a small subset of contrasts is of interest.

Trade-off

Greater universality often costs power.

Why FDR Became Necessary

In large-scale screening problems,

- FWER control may leave too few discoveries,
- especially when many hypotheses are tested and real signals are weak.

Core Shift

The goal changes from **avoiding any mistake** to **controlling the expected proportion of mistakes among discoveries**.

The Benjamini–Hochberg Procedure

Order the p -values:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}.$$

Find the largest k such that

$$p_{(k)} \leq \frac{k}{m}\alpha.$$

Reject the hypotheses corresponding to $p_{(1)}, \dots, p_{(k)}$.

How To Read the BH Threshold

The critical line

$$\frac{k}{m}\alpha$$

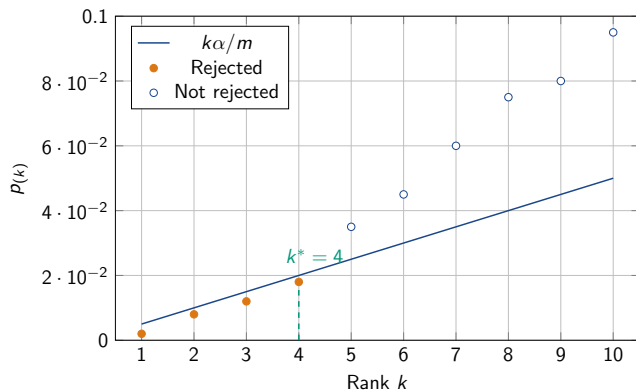
becomes less strict as the rank k increases.

- Extremely small p -values must appear first.
- Once enough small p -values accumulate, a larger block of hypotheses can be rejected.

Interpretation

BH is a **step-up selection rule** for the rejection set, not just a pointwise threshold adjustment.

BH Visualization



Reading the Plot

The largest ordered p -value below the BH line determines the rejection set.

BH Guarantee Under Independence

Let m_0 be the number of true null hypotheses. Under independence of the true-null p -values,

$$\text{FDR}_{\text{BH}} \leq \frac{m_0}{m} \alpha \leq \alpha.$$

- If all hypotheses are null, $m_0 = m$ and BH controls FDR at level α .
- If some alternatives are present, the achieved bound can be strictly smaller than α .

Positive Dependence and PRDS

BH remains valid beyond independence under suitable positive dependence.

PRDS Condition

A collection of test statistics satisfies positive regression dependence on a subset (PRDS) if increasing one coordinate tends to increase conditional probabilities of increasing events among the coordinates corresponding to true nulls.

- The formal definition is technical.
- The practical message is that BH is valid under many positively associated testing environments.

Benjamini–Yekutieli Under Arbitrary Dependence

To obtain validity under arbitrary dependence, BY replaces the BH threshold by

$$p_{(k)} \leq \frac{k}{mc(m)}\alpha, \quad c(m) = \sum_{i=1}^m \frac{1}{i}.$$

- Because $c(m) \approx \log(m) + \gamma$, the penalty grows logarithmically.
- This makes BY much more conservative than BH.

Why BY Is Conservative

- Arbitrary dependence is a strong adversarial requirement.
- To guarantee validity without structural assumptions, BY pays a substantial power cost.

Lesson

Robustness to any dependence comes with a price. Whenever credible structure is available, exploiting it can materially improve inference.

Adjusted p -values Under BH and BY

For reporting, one often computes multiplicity-adjusted values.

BH Adjusted Values

For ordered p -values,

$$\tilde{p}_{(i)} = \min_{j \geq i} \left\{ \frac{m}{j} p_{(j)} \right\} \wedge 1.$$

Analogous monotone adjustments exist for BY and stepwise FWER procedures.

Interpreting Adjusted p -values Carefully

An adjusted p -value is the smallest target level at which the hypothesis would be rejected by the specified multiplicity procedure.

- It is procedure-specific.
- It does not magically become a posterior probability of truth.
- It is useful because it separates **evidence ranking** from the final chosen target level.

A Deeper View of the BH Geometry

Consider the empirical cdf of the p -values,

$$\hat{F}_m(t) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}(p_i \leq t).$$

BH chooses the largest threshold t such that

$$\hat{F}_m(t) \geq \frac{t}{\alpha}.$$

Interpretation

BH can be viewed as intersecting the empirical p -value cdf with a linear boundary.

Storey's Perspective: The Role of π_0

Write

$$\pi_0 = \frac{m_0}{m}.$$

The BH bound is

$$\text{FDR}_{\text{BH}} \leq \pi_0 \alpha.$$

- If $\pi_0 \ll 1$, standard BH may be conservative relative to what is achievable.
- This motivates adaptive estimation of π_0 .

Estimating π_0 from the Right Tail

Under a true null, the p -value is approximately $\text{Uniform}(0, 1)$. Hence for a tuning parameter $\lambda \in [0, 1)$,

$$\hat{\pi}_0(\lambda) = \frac{\#\{p_i > \lambda\}}{m(1 - \lambda)}.$$

- If the upper tail is mostly generated by nulls, this estimates π_0 .
- Smoothing over several λ values is often used in practice.

Storey's q -value

Definition

The q -value of a test is the minimum estimated false discovery rate at which that test would enter the rejection set.

- It is the FDR analogue of a multiplicity-adjusted significance summary.
- It summarizes the tail-area error rate attached to including that hypothesis among discoveries.

p -value, Adjusted p -value, and q -value

Quantity	Interpretation
p -value	evidence against the null model for an individual test
Adjusted p -value	smallest procedure-specific target level at which the hypothesis would be rejected
q -value	smallest estimated FDR level at which the hypothesis would be included in the rejection set

None of these is the literal probability that the null is true.

Local FDR as a Two-Group Mixture Quantity

Suppose the test statistic density is modeled as

$$f(z) = \pi_0 f_0(z) + (1 - \pi_0) f_1(z).$$

Then

$$\text{lfdr}(z) = \frac{\pi_0 f_0(z)}{f(z)}.$$

- This is a posterior probability under the mixture model.
- It is more local than the tail-area quantity underlying the q -value.

Why Tail-Area and Local Notions Differ

- The q -value asks about the expected false discovery burden among all findings at least as extreme as the current one.
- The local FDR asks about the null probability at the specific observed statistic.

Practical Distinction

The first is naturally linked to thresholding a rejection set; the second is naturally linked to ranking and empirical Bayes interpretation.

Why Resampling Exists

Closed-form corrections often struggle when the joint dependence structure is complicated.

- Imaging,
- high-dimensional regression,
- correlated omics features,
- repeated-measure systems.

Idea

Use permutation or bootstrap procedures to estimate the joint null distribution of the test statistics directly from the data.

Westfall–Young MaxT Principle

For family-wise control, a natural resampling statistic is the maximum over all test statistics,

$$M = \max_{1 \leq i \leq m} T_i.$$

If the null distribution of M is estimated by permutation, one may reject H_i when

$$T_i > c_{1-\alpha},$$

where $c_{1-\alpha}$ is the $(1 - \alpha)$ quantile of the resampled distribution of M .

Why MaxT Controls FWER

By construction,

$$\mathbb{P} \left(\max_{i \in I_0} T_i > c_{1-\alpha} \right) \leq \alpha.$$

Therefore the probability that any true null statistic crosses the common resampling threshold is controlled.

- This automatically accounts for dependence among the test statistics.
- It can be much less conservative than Bonferroni when statistics are strongly correlated.

Subset Pivotality

Some exact resampling arguments rely on a condition often called subset pivotality.

Informal Meaning

The joint null distribution of the test statistics corresponding to any subset of true null hypotheses is the same whether or not false nulls are present elsewhere in the family.

This permits strong family-wise control by calibrating the procedure from the global null resampling law.

Permutation Validity Requires Design Respect

Permutation is not a generic button; it depends on exchangeability under the null.

- In randomized experiments, permutation may be exact.
- In observational settings, naive permutation may be invalid.
- Restricted permutations may be needed for blocks, strata, or paired designs.

Bootstrap-Based Multiplicity

Bootstrap methods can be used when permutation is not available.

- Resample in a way compatible with the inferential target.
- Recompute the vector of test statistics.
- Estimate critical values or adjusted p -values from the empirical joint distribution.

Caution

The validity of bootstrap calibration depends on the asymptotic regime and on how accurately the bootstrap approximates the joint law.

Single-Step and Step-Down Resampling

- **Single-step:** one common critical value from the joint null law.
- **Step-down:** recalculate thresholds after removing rejected hypotheses.

Why Step-Down Helps

Once some hypotheses are confidently rejected, the multiplicity burden on the remaining hypotheses can be reduced.

Resampling vs Analytic Corrections

Feature	Analytic rules	Resampling rules
Assumptions	often simple and explicit	often weaker but design-specific
Dependence handling	limited unless modeled	captured empirically
Computation	light	heavier
Transparency	very high	may depend on implementation details

Selective Inference and Post-Selection Testing

In modern pipelines, hypotheses are often created after a data-driven selection step.

- variable selection,
- lasso screen-then-test workflows,
- adaptive subgroup discovery.

Problem

Naive p -values ignore the randomness of the selection event and are typically anti-conservative.

Sample Splitting and Cross-Fitting as Safeguards

One way to reduce selection-induced bias is to separate discovery from confirmation.

- use one sample to screen hypotheses,
- use an independent sample to test them,
- or cross-fit to recover efficiency while preserving validity.

Trade-off

You gain validity but may sacrifice power because the data must serve two roles.

Multiplicity in Linear Models

Whenever a model yields several coefficient tests,

$$H_{0j} : \beta_j = 0, \quad j = 1, \dots, p,$$

one has a multiple testing problem if the family of inferential claims includes more than one coefficient.

- Regression output often hides the multiplicity issue behind one row per coefficient.
- Joint confidence regions and simultaneous intervals are the corresponding model-based solutions.

Simultaneous Confidence Intervals

Testing and interval estimation are dual.

Dual View

A family-wise testing procedure induces a set of simultaneous confidence intervals with joint coverage at least $1 - \alpha$.

- Bonferroni tests correspond to Bonferroni simultaneous intervals.
- Tukey and Scheffé similarly induce simultaneous interval statements for contrasts.

Confidence Bands vs Pointwise Intervals

For curves, dose-response functions, or event-time effects,

- pointwise intervals control uncertainty at one fixed location,
- simultaneous bands control uncertainty over the entire curve.

Analogy

Pointwise intervals stand to simultaneous bands as individual-test inference stands to multiple testing correction.

ANOVA Follow-Up Strategies

After a global ANOVA rejection, the question becomes which follow-up family is relevant.

- all pairwise mean differences → Tukey,
- each treatment vs control → Dunnett,
- arbitrary contrasts → Scheffé or model-based simultaneous inference.

Lesson

“Post hoc correction” should match the scientific family of contrasts, not just the existence of many means.

Multiplicity in Generalized Linear Models

The issue is not confined to Gaussian settings.

- logistic regression with many coefficients,
- Poisson or negative binomial rate ratios,
- multinomial or ordinal models with many contrasts.

Simultaneous inference can be built from Wald, likelihood-ratio, score, or resampling-based statistics.

Dose-Finding and MCP-Mod Logic

Dose-response studies often combine

- multiple candidate curve shapes,
- multiple contrasts,
- and a final model-based estimation stage.

Multiplicity Role

The first-stage multiple comparison procedure guards against choosing a dose-response shape purely by chance.

Interim Looks and Alpha Spending

Sequential monitoring introduces another form of multiplicity: repeated testing over time.

- Each interim analysis spends some fraction of the error budget.
- Group-sequential methods and alpha-spending functions formalize this allocation.

Connection

Repeated looks across time are mathematically analogous to repeated looks across hypotheses: both require explicit error budgeting.

Subgroup Analyses

Testing treatment effects in many subgroups creates severe multiplicity.

- Baseline categories,
- biomarker strata,
- age, sex, severity, region.

Practical Caution

Subgroup claims are especially vulnerable because the search space is often larger than what is finally reported.

Multiplicity in Causal Inference

Causal analyses can generate many estimands:

- average treatment effects in several populations,
- time-specific effects,
- subgroup-specific effects,
- several intervention strategies.

If several inferential claims are entertained, simultaneous error control becomes relevant even when the framework is explicitly causal.

Multiplicity in Machine Learning Evaluation

Hyperparameter search, model comparison, and repeated benchmark evaluation all create multiplicity.

- selecting the best model after many attempts,
- testing many features,
- checking many metrics.

Interpretation

Overfitting can be viewed as an implicit multiplicity problem in function space.

Weighted Hypotheses

Some hypotheses matter more than others. Weighted BH replaces the ordinary thresholds by weights w_i satisfying suitable normalization.

Idea

More prior credibility or scientific importance can justify a less severe threshold for selected hypotheses, while preserving overall FDR control.

Grouped and Hierarchical FDR

When hypotheses are naturally organized into pathways, modules, or endpoint families,

- one may test groups first,
- then test within selected groups,
- while controlling a family-level error target across the hierarchy.

Online FDR

In streaming environments, hypotheses arrive over time.

- You cannot sort all p -values in advance.
- Procedures such as alpha-investing and LORD allocate a testing budget sequentially.

Use Case

A/B platforms, surveillance systems, and adaptive experimentation pipelines.

Knockoffs

For variable selection with many correlated predictors, knockoff methods create synthetic controls that mimic the dependence structure of the original covariates.

- They target FDR control for selected variables.
- They are especially useful when the scientific object is a selected feature set rather than a fixed family of classical null hypotheses.

Empirical Bayes Perspective

Large-scale testing can be framed through mixture models and prior distributions on effects.

- Estimate the null proportion,
- estimate the effect distribution,
- derive posterior summaries such as local false discovery rates.

Bridge

Empirical Bayes does not replace multiplicity thinking; it provides another language for simultaneous shrinkage and error calibration.

Why Adaptivity Helps and Hurts

Adaptive procedures can improve power by learning from the ensemble of tests.

- estimate π_0 ,
- weight families differently,
- incorporate structure.

But every adaptive gain depends on assumptions, tuning, or stability of the learning step.

The Role of Dependence Structure

Dependence is not a nuisance detail; it fundamentally determines how conservative a generic correction must be.

- Strong positive correlation can make Bonferroni overly pessimistic.
- Negative or adversarial dependence can invalidate more aggressive rules.

Takeaway

Knowing the dependence structure is often equivalent to knowing how much power can be safely recovered.

Simultaneous Inference Is About Sets

A useful mental model is this:

- a single test asks whether one point enters the rejection region,
- a multiple testing procedure asks which **set of hypotheses** can be rejected while respecting a global error constraint.

StatHarbor View

The object of inference is the rejection set itself.

Clinical Trials

Multiplicity appears through

- multiple primary endpoints,
- multiple doses,
- interim looks,
- subgroup claims,
- secondary and exploratory families.

Strong family-wise control is often required for confirmatory labeling claims.

A Confirmatory Clinical Trial Architecture

1. Test primary endpoint family.
2. If successful, pass alpha to key secondary endpoints.
3. Optionally gate subgroup or biomarker families.

Interpretation

Multiplicity is managed not only by formulas but by **prospectively designed testing hierarchies**.

Genomics and Omics

Massive feature spaces make FDR control natural.

- thousands of genes,
- tens of thousands of proteins,
- millions of variants.

The cost of missing true signals may be high, while some controlled proportion of false leads is scientifically acceptable.

Neuroimaging

Voxel-wise testing forces one to confront both multiplicity and spatial dependence.

- naive voxel-wise thresholds produce false activation maps,
- cluster-wise and random-field methods address spatial structure,
- permutation-based max-statistic methods provide robust alternatives.

A/B Testing and Product Experiments

Modern experimentation platforms often test

- many features,
- many metrics,
- many user segments,
- over repeated deployment cycles.

Practical Choice

FDR-oriented rules may be more aligned with business value than stringent family-wise control, provided the false discovery burden remains governed.

Backtesting many strategies on historical data creates a severe multiplicity problem.

- The best-performing strategy in-sample may be selected by chance.
- Data snooping corrections are essential.

Interpretation

Uncorrected strategy search is best viewed as a multiple testing engine for generating spurious Sharpe ratios.

Epidemiology and Public Health Surveillance

Monitoring many signals across regions, outcomes, or time points can create massive surveillance families.

- safety signals,
- outbreak detection,
- many exposures across many outcomes.

Both family-wise and false discovery style criteria can be justified depending on the cost of false alarms.

Model Building in Regression

When dozens or hundreds of candidate covariates are considered, the reported coefficients inherit a selection problem.

- Pre-specification reduces multiplicity.
- Penalization reduces variance but does not automatically solve inferential multiplicity.
- Post-selection inference or holdout confirmation may still be needed.

How To Choose an Error Criterion

Goal	Natural criterion
No false confirmatory claim	FWER
Allow some controlled noise in screening	FDR
Allow a bounded number of errors	k -FWER or PFER
Rank findings by evidence under large-scale testing	q -values or local FDR

Questions To Ask Before Choosing a Procedure

1. What is the family of hypotheses?
2. Is the setting confirmatory or exploratory?
3. What kind of false positive burden is scientifically tolerable?
4. Are the tests independent, positively dependent, or strongly structured?
5. Is a specialized comparison family available?

What Should Be Reported

A transparent report should state

- the family of hypotheses considered,
- the target error criterion,
- the multiplicity procedure,
- any structural assumptions or resampling design,
- adjusted summaries such as simultaneous intervals, adjusted p -values, or q -values.

Common Reporting Mistakes

Frequent Errors

- reporting unadjusted p -values after testing many endpoints,
- choosing the correction after looking at the results,
- calling a q -value a posterior probability of truth,
- using permutation without checking exchangeability,
- claiming “exploratory” after conducting a confirmatory-style multiplicity search.

A Practical Decision Grid

Scenario		Recommended family	Typical procedure
Primary	confirmatory	small, prespecified	Holm, gatekeeping, closed testing
All pairwise means		structured contrasts	Tukey HSD
Many genomic features		very large, exploratory	BH, adaptive BH, q -values
Complex dependence		structured, correlated	Westfall–Young or bootstrap calibration

The Conceptual Shift

Old Question

“Is this one test significant?”

Multiplicity-Aware Question

“Given the whole family of candidate findings, which rejection set can be defended under an explicitly controlled error criterion?”

Final Takeaways I

- A standard p -value provides an **individual-test** guarantee, not automatic family-level control.
- Multiple testing is a problem of **simultaneous decision-making**.
- The correct response is to define the right family-level error criterion before making claims.

Final Takeaways II

- FWER is appropriate when even one false claim is costly.
- FDR is appropriate when broad discovery matters and some controlled false signal is acceptable.
- Exploiting structure, dependence, and hierarchy often improves power relative to generic worst-case corrections.

Final Takeaways III

- Adaptive procedures can recover power, but only through additional assumptions or estimation steps.
- Resampling procedures are powerful because they approximate the joint law directly.
- Reporting must make the family, criterion, and procedure explicit.

The StatHarbor Principle

Core Insight

Every additional degree of analytical freedom increases the need for explicit simultaneous error control.

Appendix: Why $\text{FDR} \leq \text{FWER}$

Because $V \leq R$ whenever $R > 0$,

$$\frac{V}{\max(R, 1)} \leq \mathbf{1}(V \geq 1).$$

Taking expectations yields

$$\text{FDR} = \mathbb{E} \left[\frac{V}{\max(R, 1)} \right] \leq \mathbb{E}[\mathbf{1}(V \geq 1)] = \mathbb{P}(V \geq 1) = \text{FWER}.$$

Appendix: Why Holm Dominates Bonferroni

If Bonferroni rejects $p_{(i)}$, then

$$p_{(i)} \leq \frac{\alpha}{m} \leq \frac{\alpha}{m - i + 1}.$$

Thus any hypothesis rejected by Bonferroni is also rejected by Holm.

Conclusion

Holm is never less powerful than Bonferroni for the same family and target level.

Appendix: Harmonic Sum in BY

The correction factor

$$c(m) = \sum_{i=1}^m \frac{1}{i}$$

satisfies

$$\log(m+1) \leq c(m) \leq 1 + \log m.$$

Therefore the BY penalty inflates roughly like $\log m$ for large m .

Appendix: Confidence-Set Duality

Suppose a family of tests has strong FWER control at level α . Then the corresponding family of non-rejected parameter values forms a simultaneous confidence set with coverage at least $1 - \alpha$.

Meaning

Multiplicity correction in testing and simultaneous coverage in interval estimation are two sides of the same inferential principle.

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